Md Rezwan Bin Mizan

Department of Mathematics, University of Houston
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Academic Appointments

- **Instructor of Record** *Fall 2025 – Present*
  Department of Mathematics, University of Houston
- **Research Assistant** *June 2024 – Present*
  Department of Mathematics, University of Houston
- **Graduate Teaching Assistant** *August 2021 – May 2024*
  Department of Mathematics, University of Houston
- **Lecturer** *January 2018 – July 2021*
  Department of Mathematics, American International University–Bangladesh (AIUB)

Education

- **PhD in Mathematics (In progress)** *Fall 2021 – Expected Summer 2026*
  University of Houston, Texas, USA
  - Advisor: Dr. Maxim Olshanskii Co-Advisor: Dr. Ilya Timofeyev
- **MS in Applied Mathematics (pending)** *Expected Fall 2025*
  University of Houston, Texas, USA
  - Graduation application submitted; degree requirements completed.
- **MS in Applied Mathematics** *2016 – 2017*
  University of Dhaka, Bangladesh
  - Thesis: Spectral Relaxation Approach on Convective Heat Transfer of Magnetic and Non-Magnetic Particles over a Stretching Cylinder with Induced Magnetic Enhancement.
- **BS in Mathematics** *2010 – 2015*
  University of Dhaka, Bangladesh
  - Project: A Study of Differential Equations in Population Biology.

Research Interests

- Reduced Order Modeling (ROM) and Tensorial ROM for nonlinear PDEs
- Data-driven and Machine Learning-based ROMs (DL-ROM, Hybrid ROM)
- Computational Fluid Dynamics (CFD) and Multiphase Flow Modeling
- Non-Newtonian and Viscoplastic Fluid Dynamics (Herschel–Bulkley flows)
- Large-scale Parametric Simulations and Scientific Computing

Publications

- **Reduced Order Modeling of Parametrized Visco-Plastic Shallow Fluid Flow.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  In preparation.

- **Tensorial Reduced-Order Models for Multi-Species Brain Tumor Simulations.**
  Asikul Islam, Md Rezwan Bin Mizan, and Andreas Mang.
  In preparation.

- **A Parametric Tensor ROM for the Shallow Water Dam Break Problem.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  Under revision in Computers & Fluids, 2025. [arXiv]

- **Parametric Reduced Order Models for the Generalized Kuramoto–Sivashinsky Equations.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev. *Under review in Applied Mathematics and Computation*, 2025. [arXiv]

- **Mathematical and Buongiorno Nanofluid Modelling to Predict the Thermal and Solutal Performance of Magneto-Bioconvective Casson Nanofluid Flowing in a Rotatory Disk.**
  Sulyman O. Salawu, MD Shamshuddin, Md Rezwan Bin Mizan, and Sajjadul Bari.
  Physica Scripta, Nov 2024.
  DOI: 10.1088/1402-4896/ad8e1a.

- **Computation of Ferromagnetic/Nonmagnetic Nanofluid Flow Over a Stretching Cylinder with Induction and Curvature Effects.**
  Md Rezwan Bin Mizan, Mohammad Ferdows, MD Shamshuddin, et al.
  Heat Transfer, 2021.
  DOI: 10.1002/htj.22122.

Conference and Seminar Talks

- **A Parametric Tensor ROM for the Shallow Water Dam-Break Problem.**
  Minisymposium, SIAM TX–LA Conference, UT Austin, September 27, 2025.
- **Tensorial Reduced Order Modeling of Viscoplastic Shallow-Water Dam-Break Flows.**
  Scientific Computing Seminar, University of Houston, September 18, 2025.
- **POD–ROMs for the Generalized Kuramoto–Sivashinsky Equation.**
  Graduate Student Paper Presentation, University of Houston, April 12 2024.

Poster Presentations

- **POD–ROMs for the Generalized Kuramoto–Sivashinsky Equation.**
  SIAM Texas–Louisiana Section Meeting, Baylor University, Oct 11–13, 2024.
- **Reduced Order Modeling of Kuramoto–Sivashinsky Equation using Proper Orthogonal Decomposition.**
  Mathematical Opportunities in Digital Twins (Math-DT), George Mason University, Dec 11–13, 2023.

Teaching Experience

- **Instructor of Record (MATH 2131: Linear Algebra with `MATLAB`)** *Aug 2025 – Present*
  University of Houston
  - Independently design and teach course integrating theory with MATLAB-based computation.
  - Develop assignments, exams, and coding activities; mentor and advise students through office hours.
- **Graduate Teaching Assistant** *Aug 2021 – May 2024*
  University of Houston
  - Led Calculus I–III labs and tutoring sessions at CASA; guided students through conceptual and problem-solving challenges.
  - Graded coursework for Discrete Mathematics, Probability, Linear Algebra, and Ordinary Differential Equations.
- **Lecturer** *Jan 2018 – Jul 2021*
  American International University–Bangladesh (AIUB)
  - Taught undergraduate courses in Calculus, Differential Equations, Complex Variables, Numerical Methods, and Linear Algebra.
  - Designed syllabi, delivered lectures, evaluated student work, and mentored students on academic progress and projects.

Leadership & Service

- **Treasurer** *2023–Present*
  AMS Student Chapter, University of Houston
- **Public Relations & Publicity Secretary** *2025–2026*
  Graduate Student Association Bangladesh-UH Executive Committee

Technical Skills

- **Programming Languages**: Python, MATLAB, FORTRAN, Mathematica
- **Languages**: English (fluent), Bengali (native)

Professional Training

- **TAMIDS Scientific Machine Learning (SciML) Summer School 2025**, Texas A&M University.
- **Workshop on Presentation, Goal Setting & Professional Development**, Life Carnival Foundation.
- **Faculty Development Webinar on OBE**, AIUB.

Honors & Fellowships

- **Research Assistantship (NSF Awards DMS-2309197, DMS-2408978)** *2024 – Present*
  University of Houston
- **Graduate Tuition Fellowship** *2021 – Present*
  University of Houston
- **Teaching Assistantship** *2021 – Present*
  University of Houston
- **Bangladesh Sweden Trust Fund** *2021*
  Ministry of Finance, Bangladesh
- **National Science and Technology Fellowship** *2016 – 2017*
  Bangladesh

Memberships

- **Member** *2022–Present*
  American Mathematical Society (AMS)
- **Member** *2023–Present*
  Society for Industrial and Applied Mathematics (SIAM)